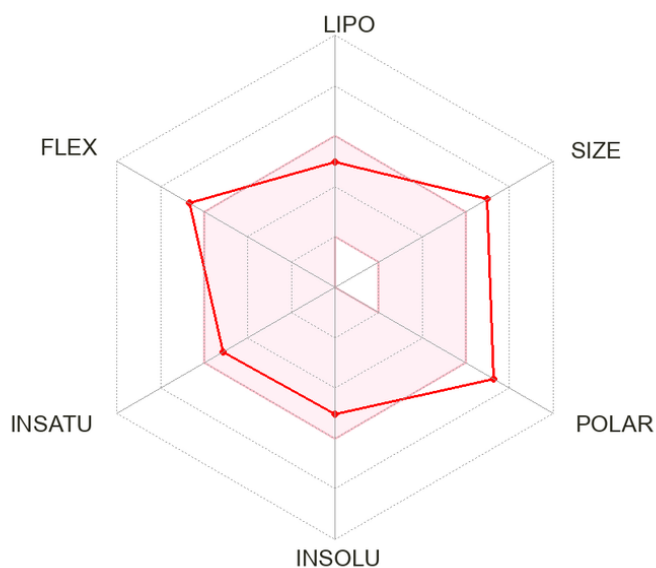
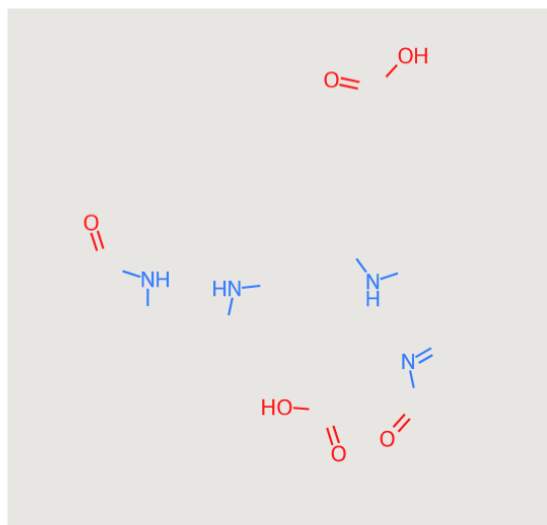


SwissADME Report

Compound: (2R,3Z)-Phycocyanobilin | **Formula:** C₃₃H₃₈N₄O₆

Source: SwissADME (swissadme.ch) — ADMET profiling, BRCA2 docking project



SMILES: CCC1=C(C(=NC1=O)/C=C\2/C(=C(/C(=C/C3=C(C(=C(N3)/C=C\4/C(=C\C)/[C@H](C(=O)N4)C)C)CCC(=O)O)/N2)CCC(=O)O)C

Physicochemical Properties

Molecular weight	586.68 g/mol
Formula	C ₃₃ H ₃₈ N ₄ O ₆
Num. heavy atoms	43
Num. arom. heavy atoms	10
Fraction Csp ³	0.36
Num. rotatable bonds	10
Num. H-bond acceptors	7
Num. H-bond donors	5
Molar Refractivity	174.09
TPSA	164.71 Å ²

Lipophilicity

Log Po/w (iLOGP)	Temporarily unavailable
Log Po/w (XLOGP3)	3.22
Log Po/w (WLOGP)	2.49
Log Po/w (MLOGP)	1.77
Log Po/w (SILICOS-IT)	4.85
Consensus Log Po/w	3.08

Water Solubility

Log S (ESOL)	-5.02
Solubility (ESOL)	5.63e-03 mg/ml ; 9.59e-06 mol/l
Class (ESOL)	Moderately soluble

Log S (Ali)	-6.35
Solubility (Ali)	2.61e-04 mg/ml ; 4.45e-07 mol/l
Class (Ali)	Poorly soluble
Log S (SILICOS-IT)	-5.56
Solubility (SILICOS-IT)	1.60e-03 mg/ml ; 2.73e-06 mol/l
Class (SILICOS-IT)	Moderately soluble

Pharmacokinetics	
GI absorption	Low
BBB permeant	No
P-gp substrate	Yes
CYP1A2 inhibitor	Temporarily unavailable
CYP2C19 inhibitor	Temporarily unavailable
CYP2C9 inhibitor	Temporarily unavailable
CYP2D6 inhibitor	Temporarily unavailable
CYP3A4 inhibitor	Temporarily unavailable
Log Kp (skin permeation)	-7.59 cm/s

Druglikeness	
Lipinski	Yes; 1 violation: MW>500
Ghose	No; 3 violations: MW>480, MR>130, #atoms>70
Veber	No; 1 violation: TPSA>140
Egan	No; 1 violation: TPSA>131.6
Muegge	No; 1 violation: TPSA>150
Bioavailability Score	0.11

Medicinal Chemistry	
PAINS	0 alert
Brenk	0 alert
Leadlikeness	No; 2 violations: MW>350, Rotors>7
Synthetic accessibility	6.11